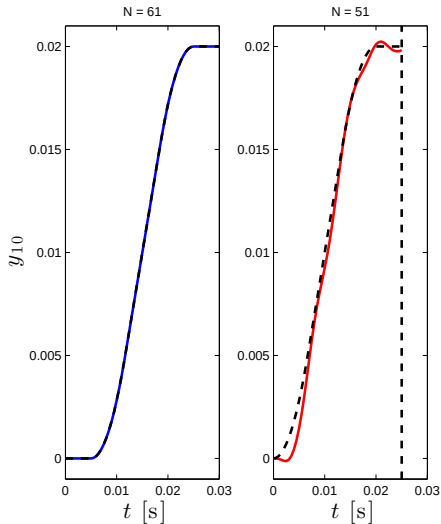


Optimization-based ILC design

Tom Oomen

Control Systems Technology Group
Department of Mechanical Engineering
Eindhoven University of Technology

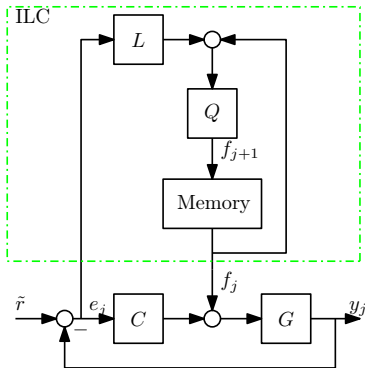
Example: frequency domain ILC



Result

- ▶ $L = (GS)^{-1}$, $Q = 1$
- ▶ faster trajectory
- ▶ 10 iterations
- ▶ trial length N
 - ▶ $N = 61$: expected result
 - ▶ $N = 51$: what's wrong?

ILC scheme



ILC summary

- ▶ learning update: $f_{j+1} = Q(f_j + Le_j)$
- ▶ frequency domain design procedure
 - ▶ $L = (GS)^{-1}$
 - ▶ $|Q(1 - GSL)| < 1, \forall \omega \in [0, 2\pi]$
- ▶ underlying assumptions
 - ▶ infinite time
 - ▶ LTI
 - ▶ SISO (mostly)
- ▶ this part:
 - ▶ finite time
 - ▶ LTI & LTV
 - ▶ optimal
 - ▶ mainly MIMO
 - ▶ basis functions
 - ▶ ...

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

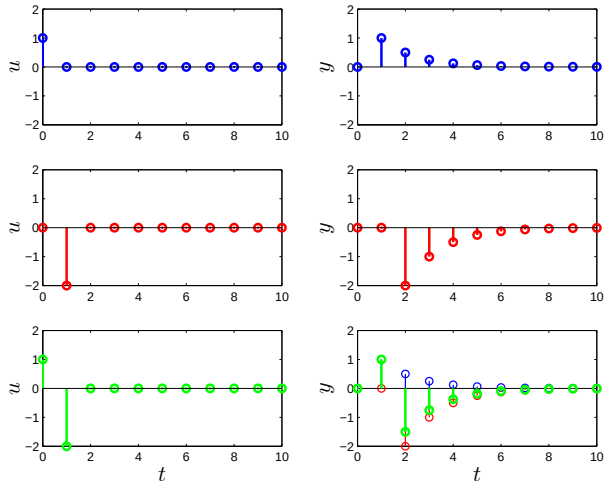
Design Procedure

Low-Order ILC

ILC for flexible tasks*

Summary

Intuitive idea



$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_{10} \end{bmatrix} u(0)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} 0 \\ h_0 \\ \vdots \\ h_9 \end{bmatrix} u(1)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} h_0 & 0 \\ h_1 & h_0 \\ \vdots & \vdots \\ h_{10} & h_9 \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \end{bmatrix}$$

Convolution causal system

$$y(k) = \sum_{l=0}^{\infty} h_l u(k-l) \quad u(k) = 0, k < 0$$

- ▶ $y(0) = h_0 u(0)$
- ▶ $y(1) = h_0 u(1) + h_1 u(0)$
- ▶ $y(2) = h_0 u(2) + h_1 u(1) + h_2 u(0)$
- ▶ $y(3) = h_0 u(3) + h_1 u(2) + h_2 u(1) + h_3 u(0)$

Convolution matrix

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \underbrace{\begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ h_3 & h_2 & h_1 & h_0 \end{bmatrix}}_{\text{convolution matrix}} \begin{bmatrix} u(0) \\ u(1) \\ u(2) \\ u(3) \end{bmatrix}$$

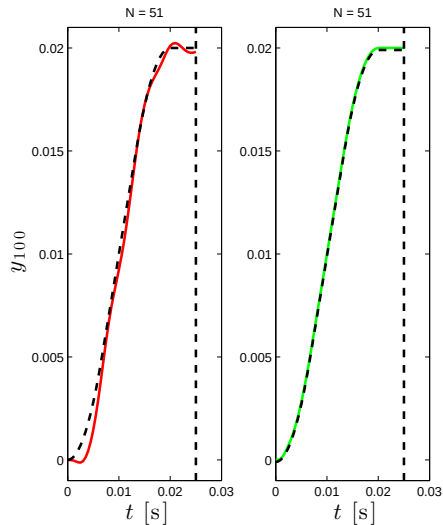
Frequency domain ILC controller

- ▶ $f_{j+1} = Q(f_j + Le_j)$
- ▶ set $L = (GS)^{-1}$
- ▶ set $Q = 1$

Intuitive lifted ILC controller

- ▶ $f_{j+1} = f_j + Le_j$
- ▶ Tikhonov regularized inverse
 - ▶ $L = (J^T J + \rho I)^{-1} J^T$
 - ▶ if J regular and $\rho = 0$, then $L = J^{-1}$
 - ▶ also known as ridge regression in statistics

Example revisited: frequency domain ILC



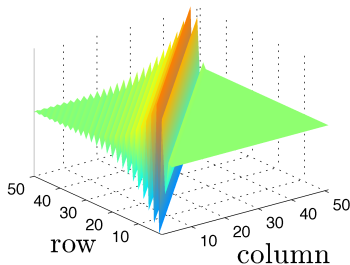
Result after many iterations

- Frequency domain: large error
- lifted ILC: small error...!

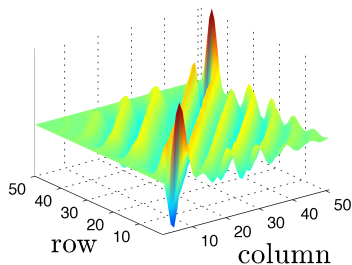
Example revisited: frequency domain ILC

Toeplitz matrix of L :

Frequency domain ILC



Lifted ILC



- Important difference: edge effects in red and blue
- Note: frequency domain ILC uses ZPETC

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

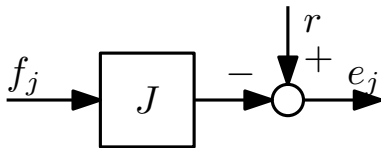
Low-Order ILC

ILC for flexible tasks*

Summary

System relations

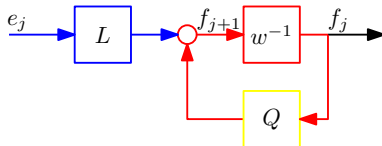
- ▶ system
 - ▶ $e_j = r - Jf_j$
 - ▶ J square, lower triangular and Toeplitz
 - ▶ time invariant
 - ▶ causal



- ▶ note: in earlier frequency domain approach $e_j = S\tilde{r} - GSf_j$
 $\Rightarrow$ take $r = S\tilde{r}$, $J = GS$

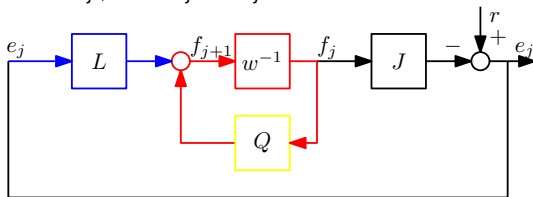
System relations

- ▶ ILC controller
 - ▶ $f_{j+1} = Qf_j + Le_j$
 - ▶ note: different definition Q compared to frequency domain ILC
 - ▶ more convenient for synthesis (Slide 21)
 - ▶ no fundamental difference - also both are used in literature
 - ▶ Q, L not necessarily lower triangular and Toeplitz
 - ▶ time varying
 - ▶ non-causality



System relations

- ▶ system $e_j = r - Jf_j$
- ▶ ILC controller $f_{j+1} = Qf_j + Le_j$



- ▶ system relations resemble state space

$$\begin{aligned} f_{j+1} &= (Q - LJ) f_j + Lr \\ e_j &= -Jf_j + r \end{aligned}$$

- ▶ system is asymptotically stable if $|\lambda_i(Q - LJ)| < 1, \quad i = 1, 2, \dots, Nl$

- recall state-space system

$$\begin{aligned} f_{j+1} &= (Q - LJ) f_j + Lr \\ e_j &= -Jf_j + r \end{aligned}$$

- state-space to iteration-domain transfer function matrix $r \mapsto e_j$

$$e_j = \left(I_{Nl} - J(wI_{Nl} - Q + LJ)^{-1} L \right) r$$

- steady-state error: set $w = 1$

$$e_{\infty} = \left(I_{Nl} - J(I_{Nl} - Q + LJ)^{-1} L \right) r$$

- let $Q = I_{Nl}$.

- then $e_{\infty} = (I_{Nl} - J(LJ)^{-1} L) r$

- $e_{\infty} = 0$ requires LJ to be non-singular $\Rightarrow J$ and L square and non-singular

Singularity of J

- ▶ reasons:
 - ▶ strictly proper systems or additional delays: if $h_0, h_1, \dots, h_{d-1} = 0$, then $\text{rank}(J)$ is at most $Nm - d$
 - ▶ non-minimum phase zeros
- ▶ consequences:
 - ▶ no full stabilization
 - ▶ no full rejection

Example

$$h = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \Rightarrow J = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \end{bmatrix}, \quad r = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Thus: $e_{\infty}(1) = 1$

Convergence properties revisited

- ▶ $e_j = r - Jf_j$
- ▶ $f_{j+1} = Qf_j + Le_j = (Q - LJ)f_j + Lr$
- ▶ asymptotically stable if $|\lambda_i(Q - LJ)| < 1, \quad i = 1, 2, \dots, N$

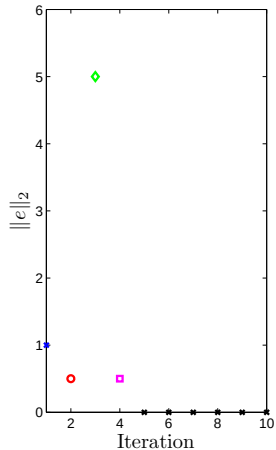
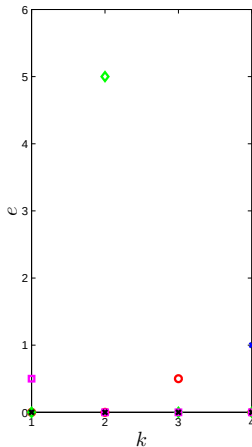
Example

$$Q = I_4, L = \begin{bmatrix} 1 & -\frac{1}{10} & 0 & 0 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}, J = I_4$$

- ▶ Q: asymptotically stable?
A: $\lambda_i(Q - LJ) = 0, \quad i = 1, 2, 3, 4 \Rightarrow$ asymptotically stable!

Example

- ▶ $f_1 = 0$
- ▶ $e_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
- ▶ error **increases** at iteration 3
- ▶ but indeed **converges** to $e_\infty = 0$
- ▶ **undesired** transient behavior



ILC learning transient

- ▶ no guarantees provided by convergence condition $\lambda_i(Q - LJ) = 0, i = 1, 2, 3, 4$
- ▶ of essential importance in practice

Result

Monotonic convergence of f_j in the ℓ_p norm, i.e.,

$$\|f_\infty - f_{j+1}\|_p \leq k \|f_\infty - f_j\|_p$$

for some $k \in [0, 1)$, is achieved if

$$\|Q - LJ\|_{p,p} < 1$$

Note: can also be defined for e_j

Convergence conditions

- ▶ convergence: $\max_i |\lambda_i(Q - LJ)| < 1$
- ▶ monotonic convergence: $\|Q - LJ\|_{p,p} < 1$

Convergence results

- ▶ spectral radius $\max_i |\lambda_i(Q - LJ)|$ is not a norm
- ▶ for any (matrix) norm $\|\cdot\|$: $\max_i |\lambda_i(Q - LJ)| \leq \|Q - LJ\|$
- ▶ hence: monotonic convergence $\Rightarrow$ convergence
- ▶ monotonic convergence in 2-norm obtained if $\bar{\sigma}(Q - LJ) < 1$
- ▶ convergence in previous part (frequency domain): also **monotonic**

Example revisited

$$\blacktriangleright L = \begin{bmatrix} 1 & -\frac{1}{10} & 0 & 0 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

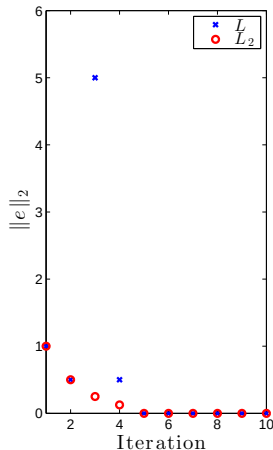
$\blacktriangleright \max_i |\lambda_i(Q - LJ)| = 0 \Rightarrow$ convergence

$\blacktriangleright \bar{\sigma}(Q - LJ) = 10 \Rightarrow$ no monotonic convergence

$$\blacktriangleright L_2 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\blacktriangleright \max_i |\lambda_i(Q - L_2J)| = 0 \Rightarrow$ convergence

$\blacktriangleright \bar{\sigma}(Q - L_2J) = \frac{1}{2} \Rightarrow$ monotonic convergence



Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

Low-Order ILC

ILC for flexible tasks*

Summary

- ▶ system relations

$$f_{j+1} = Qf_j + Le_j$$

$$e_j = r - Jf_j$$

- ▶ how to design Q and L ?
- ▶ formalization of Tikhonov regularization on slide 6
- ▶ consider criterion

$$\mathcal{J}(f_{j+1}) = e_{j+1}^T W_e e_{j+1} + f_{j+1}^T W_f f_{j+1} + (f_{j+1} - f_j)^T W_{\Delta f} (f_{j+1} - f_j)$$

- ▶ goal: compute $\arg \min_{f_{j+1}} \mathcal{J}(f_{j+1})$
- ▶ weights $W_e, W_f, W_{\Delta f}$:
 - ▶ symmetric, positive (semi) definite
 - ▶ typical choice: $W_e = I, W_{\Delta f} = w_{\Delta f} I, W_f = w_f I$, with $w_f, w_{\Delta f} \geq 0$

Reformulate

$$\mathbf{e}_j = \mathbf{r} - \mathbf{J}\mathbf{f}_j \quad \Rightarrow \quad \mathbf{r} = \mathbf{e}_j + \mathbf{J}\mathbf{f}_j$$

Hence

$$\begin{aligned}\mathbf{e}_{j+1} &= \mathbf{r} - \mathbf{J}\mathbf{f}_{j+1} \\ &= \mathbf{e}_j - \mathbf{J}(\mathbf{f}_{j+1} - \mathbf{f}_j)\end{aligned}$$

and

$$\begin{aligned}\mathcal{J}(\mathbf{f}_{j+1}) &= (\mathbf{e}_j - \mathbf{J}(\mathbf{f}_{j+1} - \mathbf{f}_j))^T \mathbf{W}_e (\mathbf{e}_j - \mathbf{J}(\mathbf{f}_{j+1} - \mathbf{f}_j)) \\ &\quad + \mathbf{f}_{j+1}^T \mathbf{W}_f \mathbf{f}_{j+1} + (\mathbf{f}_{j+1} - \mathbf{f}_j)^T \mathbf{W}_{\Delta f} (\mathbf{f}_{j+1} - \mathbf{f}_j)\end{aligned}$$

Optimal ILC

- ▶ first order necessary condition for optimality: $\frac{\partial \mathcal{J}(f_{j+1})}{\partial f_{j+1}} = 0$

Result

- ▶ use identity $\frac{\partial (s-Ax)^T W (s-Ax)}{\partial x} = -2A^T W (s-Ax)$
- ▶ then $\frac{\partial \mathcal{J}(f_{j+1})}{\partial f_{j+1}} = 0 \Rightarrow$ implies

$$(J^T W_e J + W_f + W_{\Delta f}) f_{j+1} - (J^T W_e J + W_{\Delta f}) f_j - J^T W_e e_j = 0$$

$$f_{j+1} = \underbrace{(J^T W_e J + W_f + W_{\Delta f})^{-1} (J^T W_e J + W_{\Delta f})}_Q f_j + \underbrace{(J^T W_e J + W_f + W_{\Delta f})^{-1} J^T W_e}_L e_j$$

Analysis of optimal ILC

- monotonic convergence:

$$\bar{\sigma}(Q - LJ) = \bar{\sigma}((J^T W_e J + W_f + W_{\Delta f})^{-1} W_{\Delta f}) < 1$$

- result: since $J^T W_e J + W_f \succeq 0$

1. if $J^T W_e J + W_f \succ 0$, i.e., full rank, then guaranteed monotonic convergence
2. if J singular, then set $W_f \succ 0$ to guarantee monotonic convergence
3. if $W_{\Delta f} = 0$, then deadbeat ILC
4. if J non-singular, then $W_f = W_{\Delta f} = 0$ and $W_e = w_e I$ leads to

$$f_{j+1} = f_j + (J^T J)^{-1} J^T e_j = f_j + J^{-1} e_j \Rightarrow \text{inverse model ILC}$$

Steady state error

$$e_{\infty} = \left(I_{Nl} - J(I_{Nl} - Q + LJ)^{-1}L \right) r = \left(I_{Nl} - J(J^T W_e J + W_f)^{-1} J^T W_e \right) r$$

Design guidelines

- ▶ if J non-singular, then $W_f = 0$ leads to $e_{\infty} = 0$
- ▶ can be shown that $W_f \succ 0$ leads to improved robustness w.r.t. model uncertainty
- ▶ $W_{\Delta f}$:
 - ▶ coincides with specific Tikhonov regularization on slide 6
 - ▶ large: attenuate trial-variant disturbances
 - ▶ small: faster convergence in trial-invariant disturbance situation

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

Low-Order ILC

ILC for flexible tasks*

Summary

1. Obtain an impulse response of system
 - ▶ from a given parametric model
 - ▶ from experiments
 - ▶ impulse response testing
 - ▶ estimation of FIR model using general inputs (fairly easy)
2. Compute Toeplitz matrix J
3. Select tuning parameters W_e , W_f , and $W_{\Delta f}$
4. Compute optimal ILC: L and Q on slide 21

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

Low-Order ILC

ILC for flexible tasks*

Summary

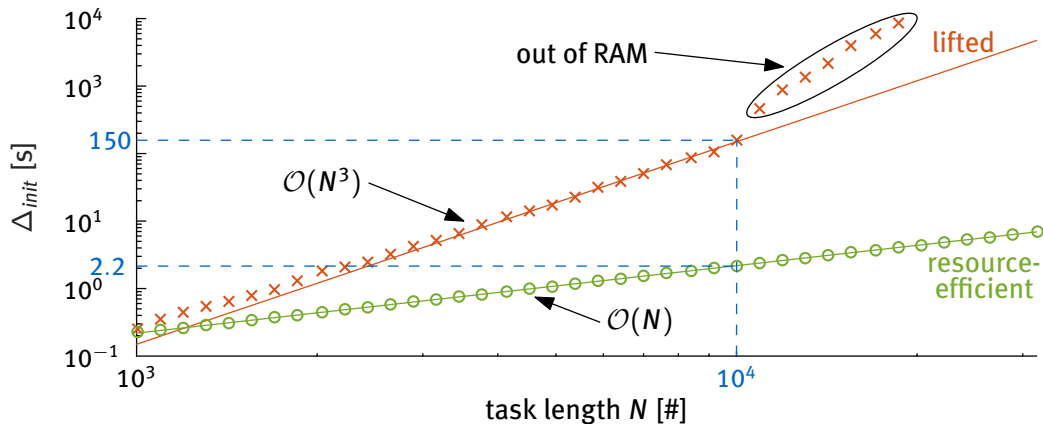
Low-order implementation of optimal ILC controller of slide 21 (Oomen et al.2011)

(van Zundert et al.2016)

$$f_{j+1} = C^{\text{ILC}}(k) \begin{bmatrix} e_j(t) \\ f_j(k) \\ g_j(k+1) \end{bmatrix}$$

- ▶ state-space $C^{\text{ILC}}(k)$ follows from Riccati difference equation
- ▶ related to LQ tracking problem: $g_{<k>}(k+1)$ noncausal
 - ▶ generalization of (Amann et al.1996)
- ▶ **tractable** for large scale problems
 - ▶ matrix dimensions invariant under
 - ▶ intersample behavior
 - ▶ $\#n_u, \#n_y$: MIMO
 - ▶ N : trial length
 - ▶ **in contrast**: approach on slide 21 **inflates** matrix dimensions

Low-order implementation of slide 21 (Oomen et al.2011) (van Zundert et al.2016)



Key point: computationally faster ILC implementation with identical control performance

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

Low-Order ILC

ILC for flexible tasks*

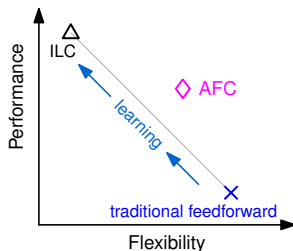
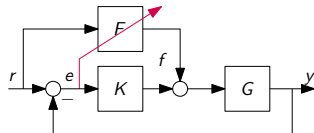
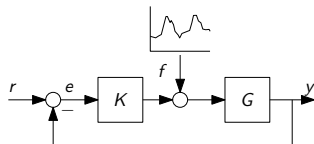
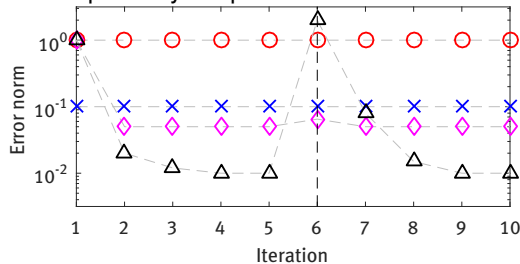
Summary

Flexibility in ILC

- ▶ Traditional ILC is inflexible to varying references...
...since it iteratively learns a signal f

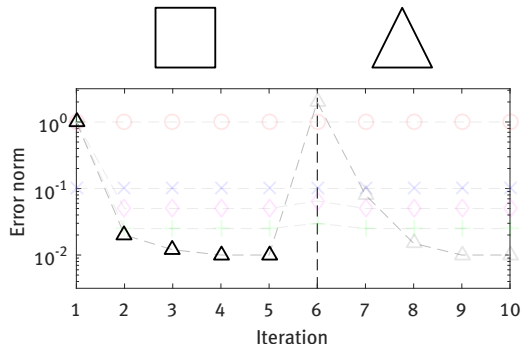
Advanced feedforward control

- ▶ Iteratively learn controller $F(\theta) = \theta\Psi$,
e.g., $\Psi = \frac{d^2}{dt^2}$ (mass feedforward)
- ▶ fits perfectly in optimization-based framework



Advanced feedforward control - drawing conclusions

Why isn't ILC standard in industrial control?



- ▶ ILC: high performance through learning, but inflexible!
- ▶ AFC: high flexibility, while maintaining good performance
- ▶ can we further improve performance of AFC?

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

Low-Order ILC

ILC for flexible tasks*

Summary

Lifted ILC

- ▶ finite time, LTI/LTV, MIMO
- ▶ general framework for repetitive control and ILC (de Roover et al.2000)

Analysis using linear algebra

- ▶ convergence vs. monotonic convergence

Synthesis

- ▶ optimal ILC algorithms

Extensions

- ▶ many different extensions and applications
 - ▶ see overview papers, e.g., (Bristow et al.2006), (Blanken et al.2016)
 - ▶ many contributions of CST M.Sc./Ph.D. students, e.g., <http://www.toomen.eu>
 - ▶ https://youtu.be/kj_ouy1Fnko?feature=shared for task-flexible ILC

- Amann, N., Owens, D. H. & Rogers, E. (1996), 'Iterative learning control for discrete-time systems with exponential rate of convergence', *IEE Proceedings Control Theory & Applications* **143**(2), 217–224.
- Blanken, L., de Rozario, R., van Zundert, J., Koekebakker, S., Steinbuch, M. & Oomen, T. (2016), Advanced feedforward and learning control for mechatronic systems, in 'Proceedings of the 3rd DSPE Conference on Precision Mechatronics', Sint-Michielsgestel, The Netherlands, pp. 79–86.
- Bristow, D. A., Tharayil, M. & Alleyne, A. G. (2006), 'A survey of iterative learning control: A learning-based method for high-performance tracking control', *IEEE Control Systems Magazine* **26**(3), 96–114.
- Oomen, T., van de Wijdeven, J. & Bosgra, O. (2011), 'System identification and low-order optimal control of intersample behavior in ILC', *IEEE Transactions on Automatic Control* **56**(11), 2734–2739.
- de Roover, D., Bosgra, O. H. & Steinbuch, M. (2000), 'Internal-model-based design of repetitive and iterative learning controllers for linear multivariable systems', *International Journal of Control* **73**(10), 914–929.
- van Zundert, J., Bolder, J., Koekebakker, S. & Oomen, T. (2016), 'Resource-efficient ILC for LTI/LTV systems through LQ tracking and stable inversion: Enabling large tasks on a position-dependent industrial printer', *Mechatronics* **38**, 76–90.